BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

LESSON 193 · Signed binary and shifts

Name: _____ Class: _____ Date: _____

Time: _____ Teacher: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. Does changing a by-value int parameter change the caller's int?

Q2. What is a return value?

Q3. Convert 00001010 from unsigned binary to decimal.

Q4. Calculate 00001111 + 00000001 using 8 bits.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

I Do · Read new knowledge

Key vocabulary

two's complement — Fixed-width signed binary representation.

sign bit — Leftmost bit indicating sign in two's complement.

logical shift — Shift with zero bits entering vacated positions.

arithmetic right shift — Shift right while extending the sign bit.

overflow — Result cannot be represented in the available bit width.

New learning

POSITIVE 13 — +13 written with 8 bits is 00001101.

INVERT BITS — Invert each bit to make 11110010.

ADD ONE — Add one to get 11110011, the 8-bit two's-complement representation of −13.

Worked example

Find −9 in 8-bit two's complement.

1. Predict first — Write +9 in 8 bits and invert all the bits.

2. Model the result — Add 1 to the inverted bits: 11110111 represents −9.

We Do · Practise with guidance

Convert −6 into 8-bit two's complement.

```
+6  = 00000110
NOT = ?
+1  = ?
```

Predict first

+6 = 00000110. Invert to get 11111001.

Model the result

Add 1: 11111010 is the representation of −6.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Question 1

Write −5 as 8-bit two's complement.

Question 2

Unsigned 8-bit 00110100 is shifted logically right one place. Give the result and decimal value.

Question 3

11110100 represents −12 in two's complement. Arithmetically shift right one place and explain the new left bit.

Exit Ticket · Answer before checking

Arithmetic right shift of a negative two's-complement value inserts zero at the left.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. No. It is a copy.

Q2. A result a function passes to its caller.

Q3. 10.

Q4. 00010000, decimal 16.

You Do pages · Independent questions

Q1. +5=00000101; inverted=11111010; add 1=11111011.

Q2. 00011010, which is 26 (52 / 2).

Q3. 11111010, representing −6. The sign bit 1 is copied into the new leftmost position.

Exit Ticket · Model answer

False. An arithmetic right shift replicates the sign bit (1 for negative values).

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question:

2. Write a complete new answer in GREEN PEN. Show working if needed: